

# Alexander Bernstein

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## Education

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**University of California, Santa Barbara**

Santa Barbara, CA

*Ph.D. in Statistics*

*Sept. 2025*

*Dissertation: “Long-Only Minimum Variance Portfolio Composition for Factor Models”*

**Washington University in St. Louis**

St. Louis, MO

*M.Sc. in Systems Science and Applied Mathematics*

*Dec. 2016*

*B.A. in Mathematics and Economics (Cum Laude)*

*May 2014*

## Research Interests

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Convex and portfolio optimization, factor models and feature construction, asset pricing (options and bonds), multivariate statistics, time-series analysis, probability and stochastic analysis, and applications of machine learning in financial mathematics.

## Publications and Working Papers

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Banerjee, T., **Bernstein, A.**, Feinstein, Z., (2025). “Dynamic clearing and contagion in financial networks”. *European Journal of Operational Research* 321.2, pp. 664–675.

**Bernstein, A.**, Shkolnik, A., (2025). “Asymptotics of Quadratic Forms on a Simplex”. In Preparation.

## Talks and Presentations

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**Analytical Solutions to the Constrained Markowitz Problem via Fixed Point Theory**

*INFORMS Annual Meeting, Phoenix, AZ*

*Oct. 2023*

**Explicit Solutions for Position Constrained Minimum Variance Portfolios**

*SIAM Conf. on Applied and Computational Discrete Algorithms, Online Poster*

*July 2021*

*CDAR Risk Seminar, UC Berkeley, Berkeley, CA*

*March 2020*

## Projects

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*Mean Field Game Reinforcement Learning:*  Implemented a **Q-Learning** algorithm to solve the major-minor cyber-security game (Carmona, Delarue, et al. 2018)[Ex. 7.2.3]. Learned optimal defense policies through interaction with a dynamic population distribution  $\mu$ .

*Conditional Covariance Estimation:*  Built a **Covariance Regression** model (**CovRegpy**) for covariate-adjusted covariance of idiosyncratic equity returns, a parsimonious alternative to **MGARCH**. Leveraged exogenous covariates to improve stability and reduce parameter overhead in high-dimensional settings.

*Quantitative Options Pricing:*  Implemented **Monte Carlo** (Euler/Milstein) and **finite-difference** (implicit, explicit, and Crank-Nicolson) solvers for the **Heston** and **CEV** models. Built **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates to recover prices and Greeks (via autograd) at lower cost.

*Factor-Model Portfolio Pipeline:* Spec-driven **Python** pipeline for S&P 500 portfolio construction. Builds **PCA** factor models with pluggable covariance estimators (**POET** thresholding, eigenvector shrinkage, **GARCH**) and return estimators (**Kalman**/EKF filters, robust Student-t, Black-Litterman, L1-adaptive control). Solves constrained mean-variance optimization via **CVXPY** (CLARABEL) with a walk-forward out-of-sample backtester and a live rebalancer.

# Teaching

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**University of California, Santa Barbara**

Santa Barbara, CA

*Teaching Assistant*

*Sept. 2017 – March 2025*

- PSTAT 173: Risk Theory · PSTAT 170: Introduction to Financial Mathematics
- PSTAT 160AB: Applied Stochastic Processes · PSTAT 174/274: Time Series Analysis
- PSTAT 126: Regression Analysis · PSTAT 120AB: Probability and Mathematical Statistics
- PSTAT 5A/109: Introduction to Statistics

*Graduate Student Mentor*

*Dec. 2019 – June 2024*

- Supervised undergraduate projects in Financial Mathematics and Optimization.
- Guided students in preparing posters and reports.

## Professional Experience

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**Epic Systems**

Madison, WI

*Technical Services Engineer*

*Sept. 2014 – Sept. 2015*

- Supported customers in the use of their Electronic Medical Records system.
- Diagnosed customer requests and tailored software to fit their needs.
- Developed and implemented improvements to the Epic codebase.

**Prozess Technologie**

St. Louis, MO

*Intern*

*May 2013 – May 2014*

- Created computational simulations to explore the effectiveness of laboratory equipment.
- Tested, calibrated, and validated laboratory equipment used in pharmaceutical manufacturing.

## Fellowships and Awards

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Regents Fellowship, *UC Santa Barbara*

*2017–2018*

John M. Olin Prize for Excellence in Economics, *Washington University in St. Louis*

*2014*

## Skills

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*Technical Knowledge*

- Expertise in Convex Optimization and Statistical Analysis.
- Strong knowledge of Mathematical Statistics, Probability, Stochastic Analysis, Machine Learning, Time Series Analysis, Data Science, Options Pricing, and Risk Analysis.

*Programming Languages and Software*

- Strong Knowledge: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), R, MATLAB,  $\text{\LaTeX}$ , Markdown, Linux.
- Working Knowledge: C, Java, JavaScript, NodeJS, SQL, MongoDB, AWS, Git, Intersystems Caché.